

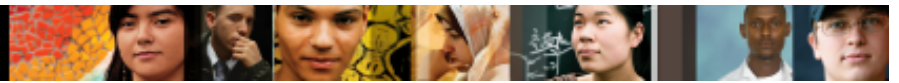


TNE10006/TNE60006: Networks and Switching



LAN Design

Cisco | Networking Academy®
Mind Wide Open™



Outline

- Hierarchical Network Model
- Converged Networks
- Network Design Considerations
- Switch Performance Characteristics
- Switch Features



Hierarchical Network Model

- To satisfy the needs of a small or medium-sized business, your network is more likely to be successful if a **hierarchical design model** is used:

This involves dividing the network into **discrete** layers

Each layer provides specific functions that define its role within the overall network

By separating the various functions that exist on a network, the network design becomes modular, which facilitates scalability and performance

- The typical hierarchical design model is broken up in to three layers:

Access

Distribution

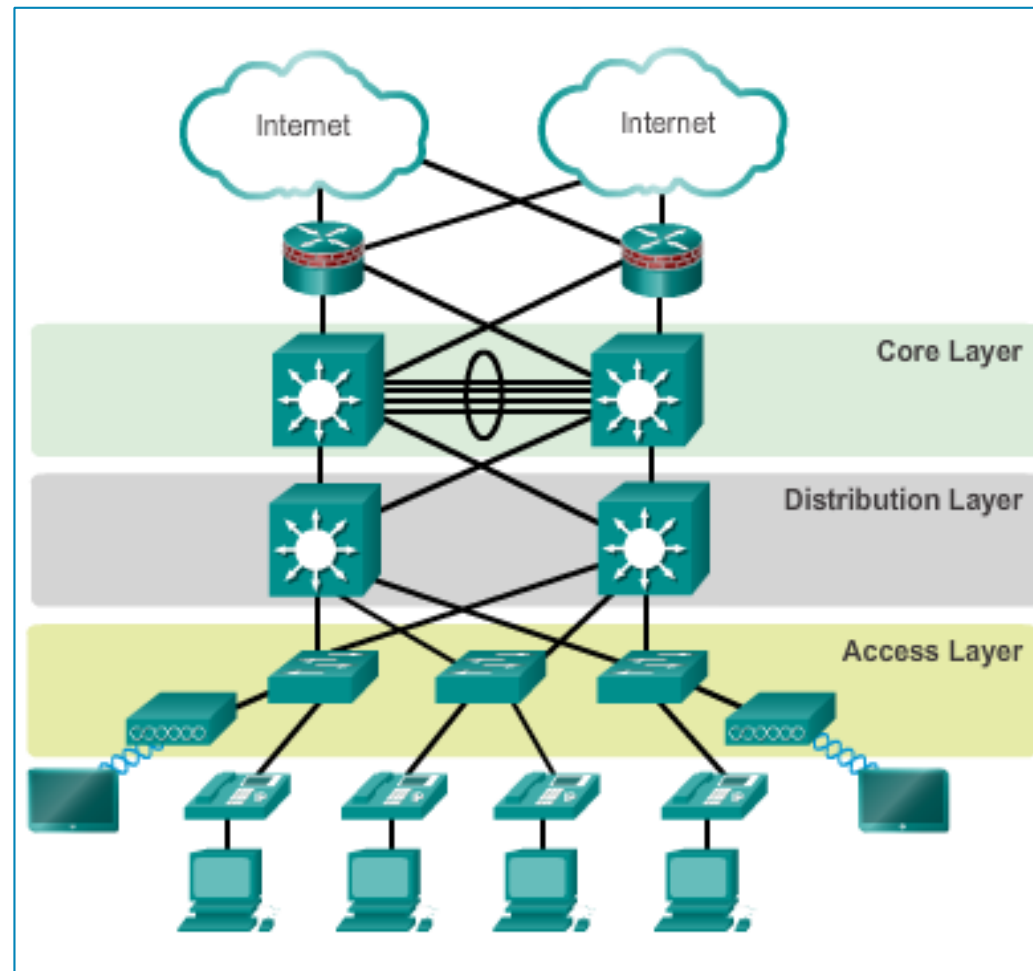
Core



Hierarchical Network Model

Hierarchical Network Design

This model divides the network functionality into three distinct layers.





Hierarchical Network Model

Access Layer

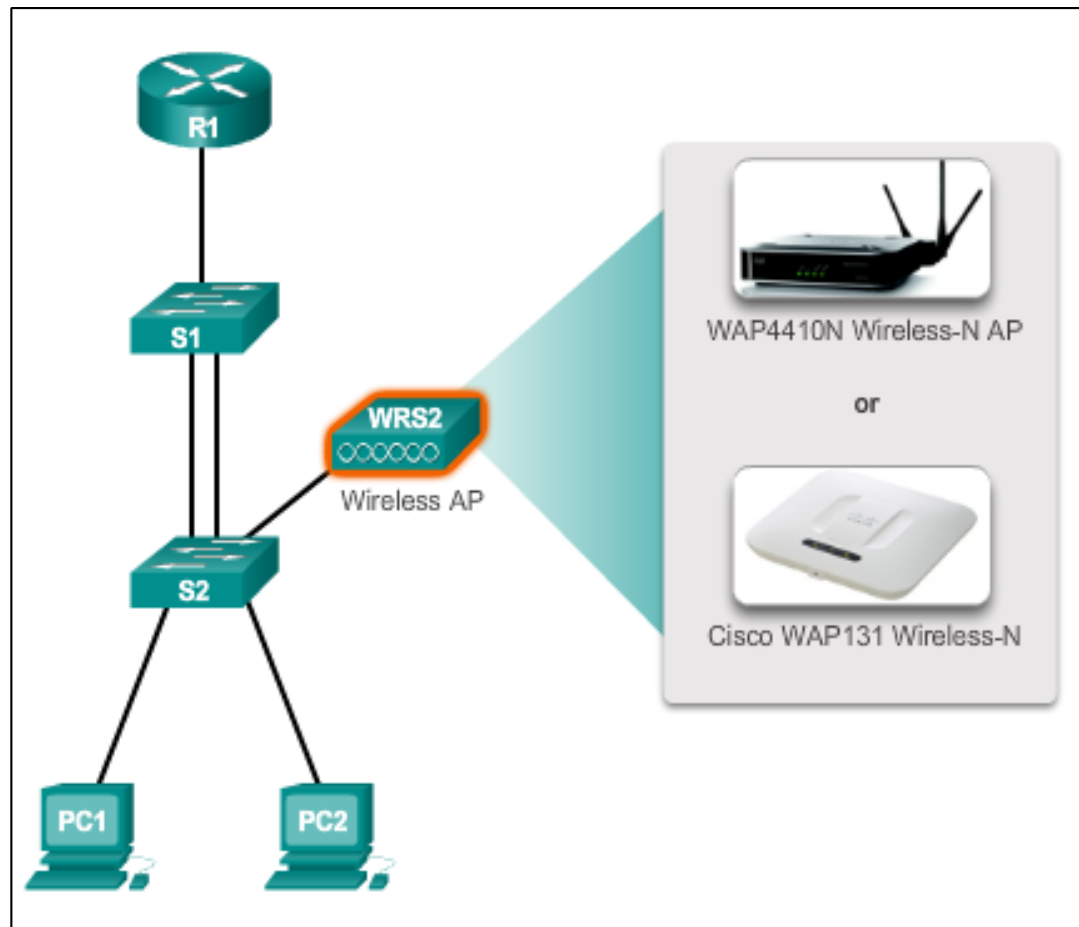
- **Interfaces with end devices**, such as PCs, printers, and IP phones, to provide **access** (2960 layer 2 switch) to the rest of the network
- Typically includes switches and wireless access points.
- The **main purpose** of this layer is to:
 - Provide a means of connecting devices to the network
 - Control which devices are allowed (eg port security) to communicate on the network



Hierarchical Network Model

Expanding the Access Layer

Access layer connectivity can be extended through wireless connectivity.





Hierarchical Network Model

Distribution Layer

- **Aggregates** the data received from the **access** layer switches before it is transmitted to the **core** layer for **routing** to its final destination
- Controls the flow of network traffic using policies (ACLs) and delineates broadcast domains by performing routing functions (3560 layer 3 switch) between **virtual LANs (VLANs) defined at the access layer**
- VLANs allow you to segment the traffic on a switch into **separate subnetworks**
- Distribution layer switches are typically high-performance devices that have high availability and redundancy to ensure reliability



Hierarchical Network Model

Core Layer

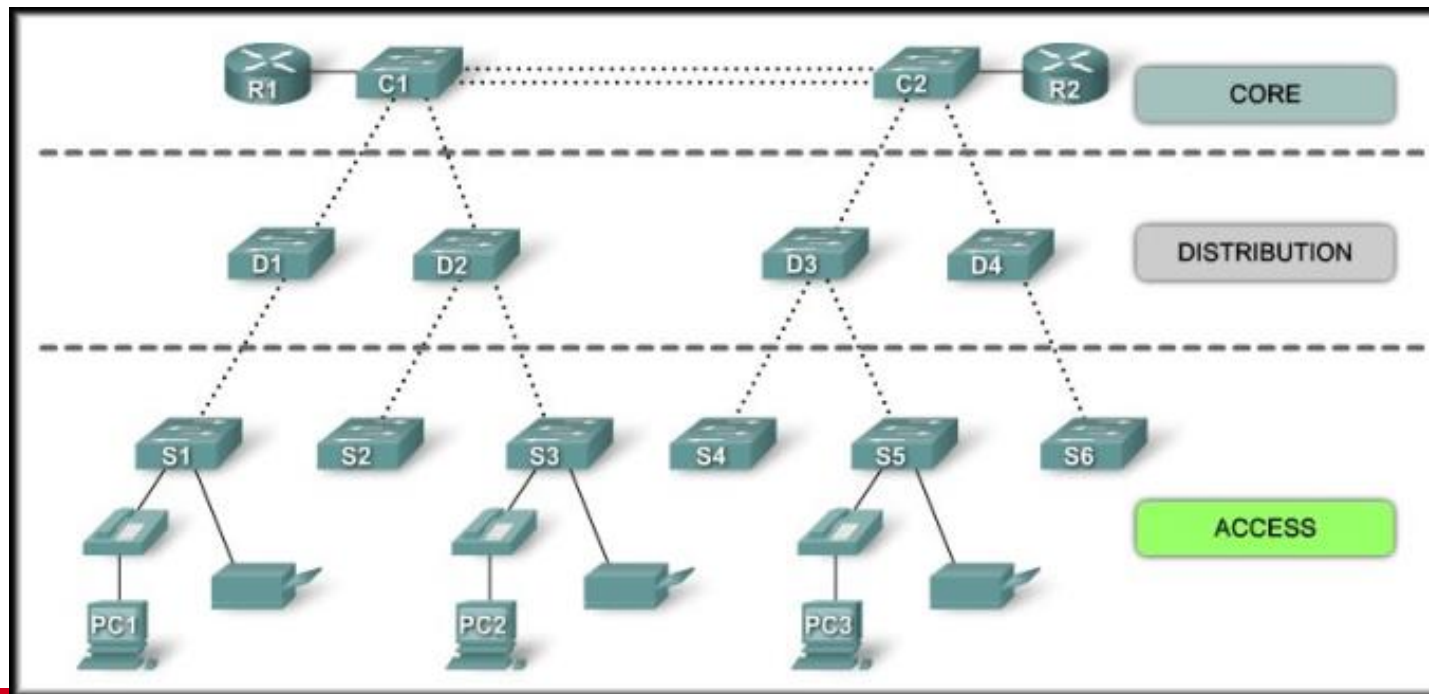
- **High-speed backbone** of the internetwork
- Critical for interconnectivity between distribution layer devices, so it is important for the core to be highly available and redundant
- The core layer can also connect to Internet resources
- Aggregates the traffic from all the distribution layer devices, so it must be capable of forwarding large amounts of data quickly
- In smaller networks, it is not unusual to implement a **collapsed model**, where the **distribution** layer and **core** layer are **combined** into one layer



Hierarchical Network Model

Logical Layout

- Layers separated into well-defined hierarchy
- Easy to see which switches perform which function
- More difficult to see in a physical/installed hierarchy
- NO intra-Layer connections



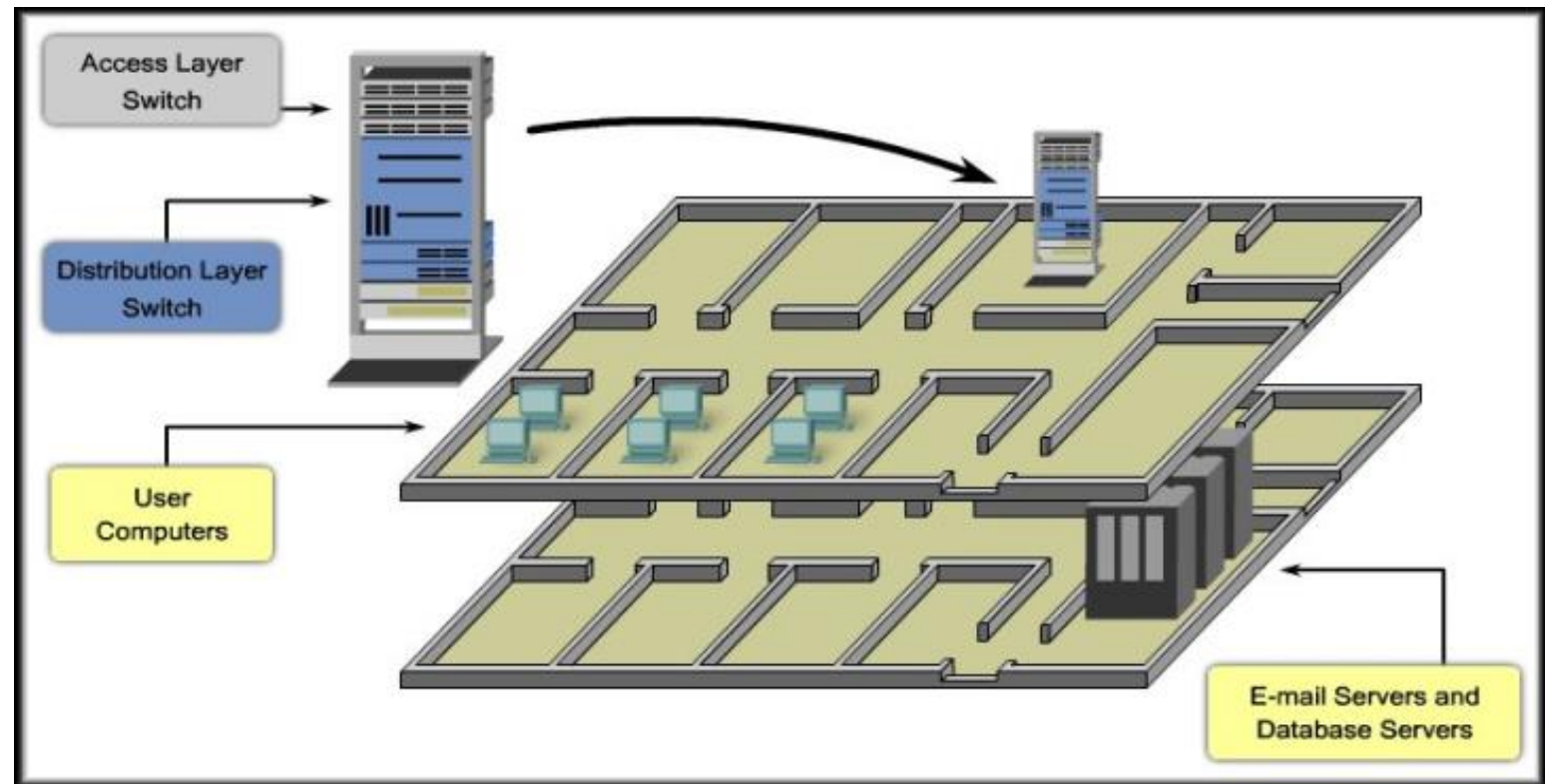


Hierarchical Network Model

Physical Layout

- Try to maintain visibility into network design
- Often access layer and distribution switches are installed in the wiring closets of each floor and connected to each of the devices needing network access
- Logical layer switches often stacked one on top of each other in the wiring closet

Physical Layout





Hierarchical Network Model

Benefits

- Scalability
- Redundancy
- Performance
 - Link aggregation between levels can allow for near wire-speed transfers throughout the network
- Security
 - How and where to implement security make installation simpler
- Manageability
 - Consistency between switch roles at each level simplifies management
- Maintainability
 - Modularity allows network to scale without extra complexity



Hierarchical Network Model

Design Principles

Network Diameter

- Number of **devices** a packet has to cross before reaching its destination
- A low diameter ensures **low** and **predictable latency** between devices

Bandwidth Aggregation

- Practice of considering the specific bandwidth requirements of each part of the hierarchy.
- After bandwidth requirements of the network are known, links between specific switches can be aggregated, which is called link aggregation

Redundancy

- Redundancy is one part of creating a highly available network.

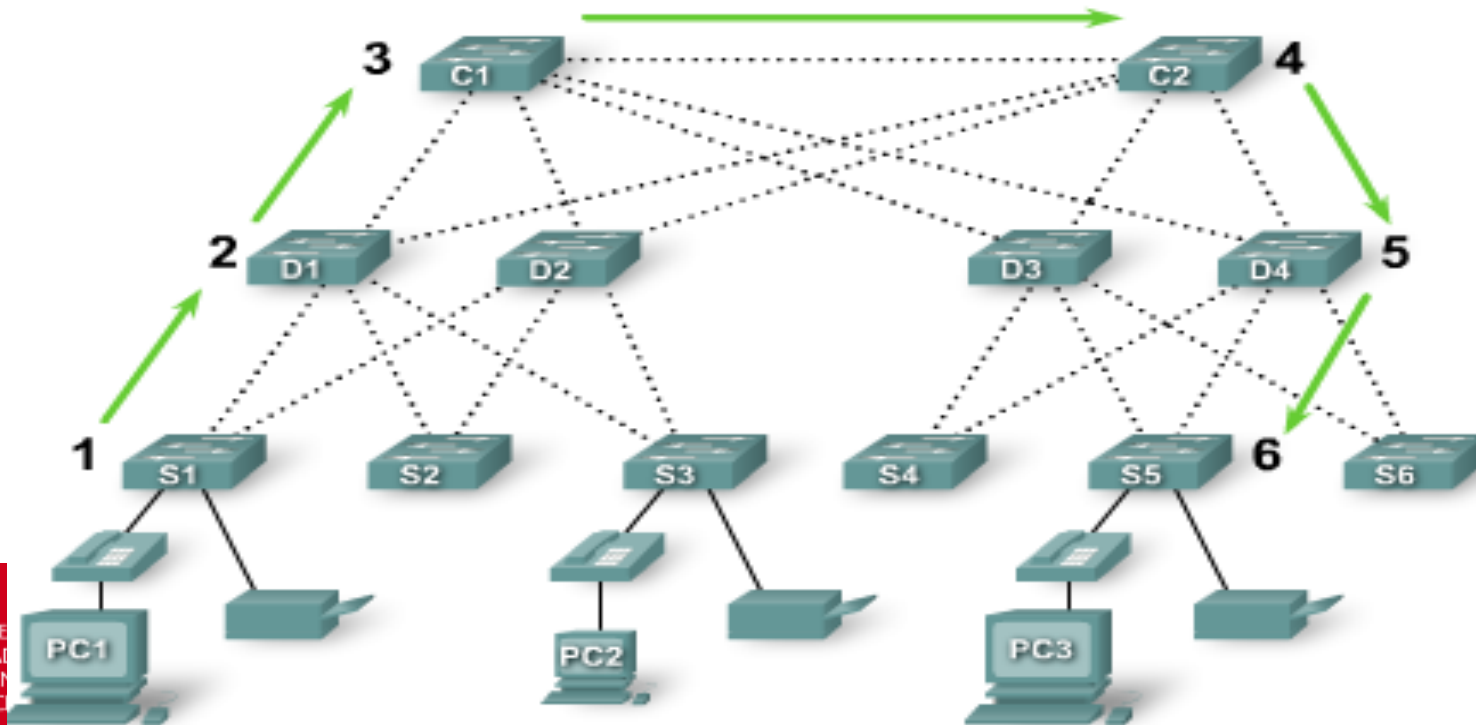


Hierarchical Network Model

Network Diameter

- In the figure, PC1 communicates with PC3. There could be up to **six interconnected switches between PC1 and PC3**. In this case, the network diameter is 6
- Each switch in the path introduces some **latency**
 - Each switch has to examine the destination MAC address, check its MAC address table, and forward the frame
 - Even though this happens in a fraction of a second, the time adds up when the frame has to cross many switches.
- In a hierarchical network, network diameter is always going to be a predictable number

Network diameter is the number of switches in the path of traffic between two endpoints.



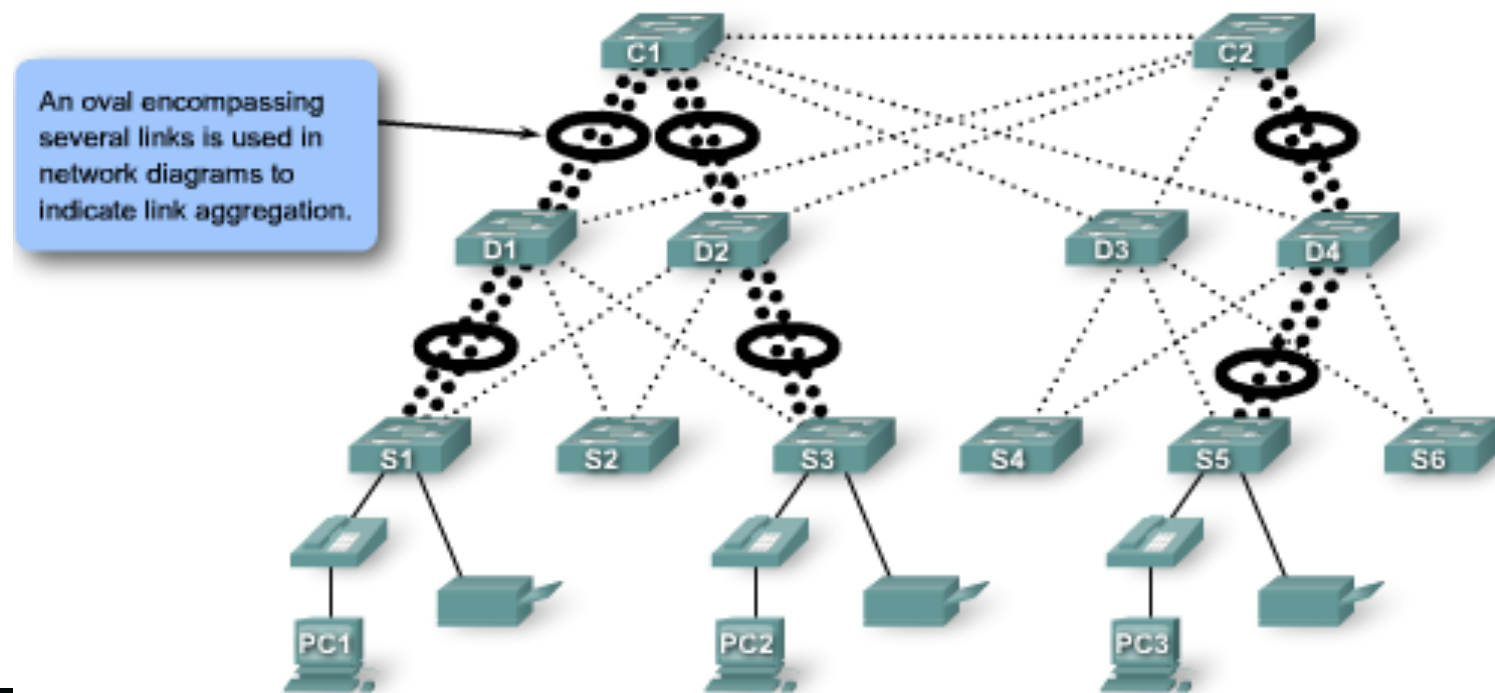


Hierarchical Network Model

Bandwidth Aggregation

- Cisco **EtherChannel** allows multiple Ethernet links to be aggregated
- Specific links on specific ports in each switch are aggregated, increasing bandwidth to targeted parts of the network

Bandwidth aggregation is normally implemented by combining several parallel links between two switches into one logical link.



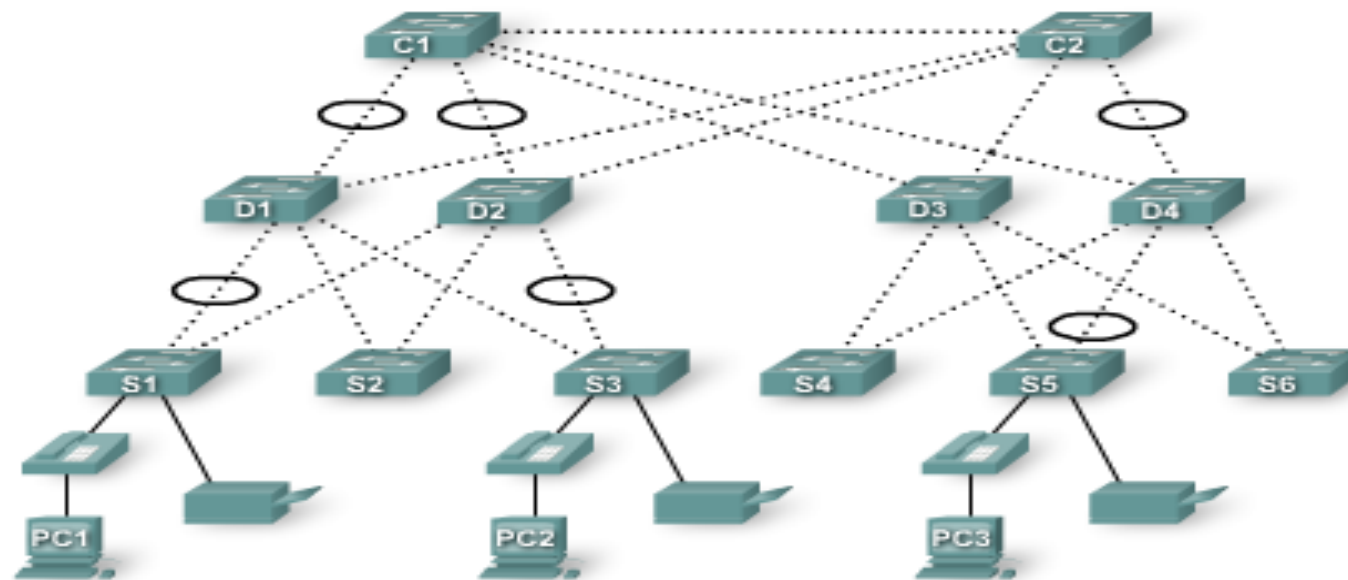


Hierarchical Network Model

Redundancy

- Redundancy can be provided in a number of ways:
 - Double the network connections between devices
 - Double the devices
- Implementing redundant links can be expensive
 - Unlikely to implement redundancy at the access layer

Hierarchical Network Design Principles





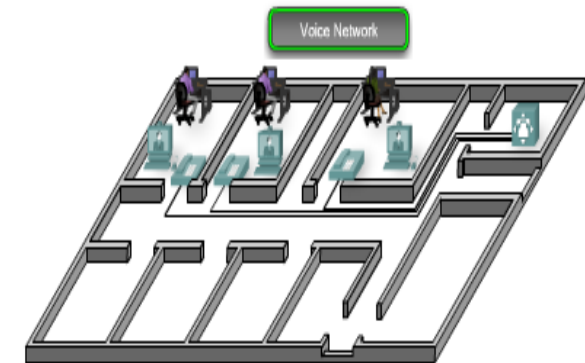
Converged Networks

Separate vs Converged Networks

Separate Voice, Video and Data Networks

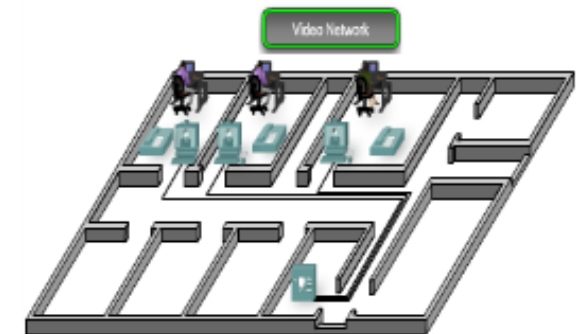
Voice network

- A voice network contains isolated phone lines running to a PBX switch to allow phone connectivity to the PSTN
- For each new phone, a new line has to be run to the PBX



Video network

- Videoconferencing data can consume significant bandwidth
- Video networks maintained separately to allow operation without bandwidth competition



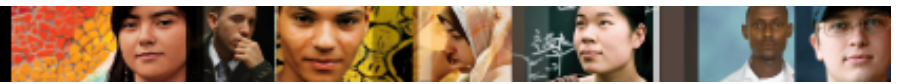
Data network

- Interconnects the workstations and servers on a network



Converged Network

- Using a properly designed hierarchical network, and implementing QoS policies that can prioritize the audio and video, means they be converged onto the data network with little to no impact on quality of service.



Switch Performance Characteristics

Port Density

- Port density is the number of ports available on a single switch

24 Port

48 Port



Very high density.
Catalyst 6500 - 1,000 Ports



Switch Performance Characteristics

Forwarding Rate

- Defines the processing capabilities of a switch by **rating how much data the switch can process per second**.
- If the switch forwarding rate is too low, it cannot accommodate full wire-speed communication across all of its switch ports
- **Access** layer switches typically do not need to operate at full wire speed
- Allows the use of:
 - Less expensive, lower performing switches at the **access layer**
 - More expensive, higher performing switches at the **distribution and core layers**, where the forwarding rate makes a bigger difference

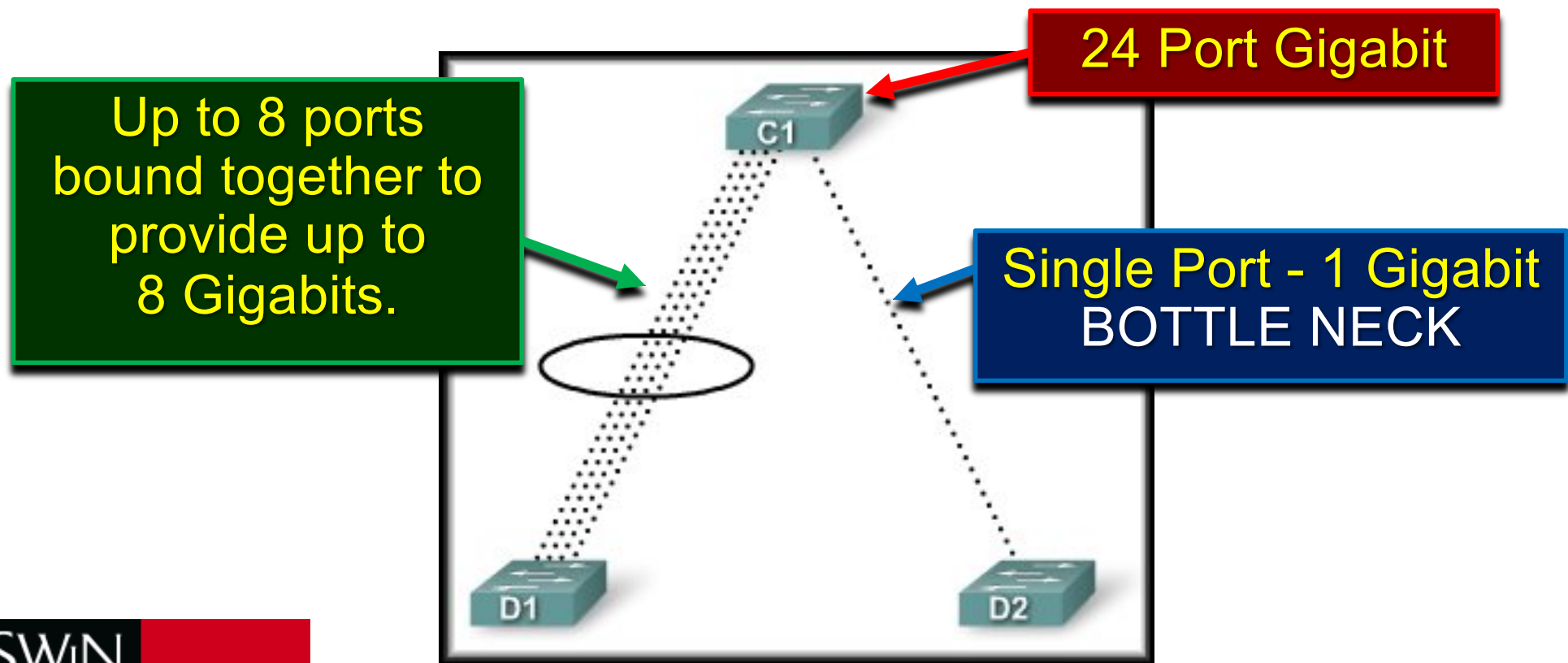




Switch Performance Characteristics

Link Aggregation

- As part of bandwidth aggregation, you should determine if there are enough ports on a switch to aggregate to support the required bandwidth.

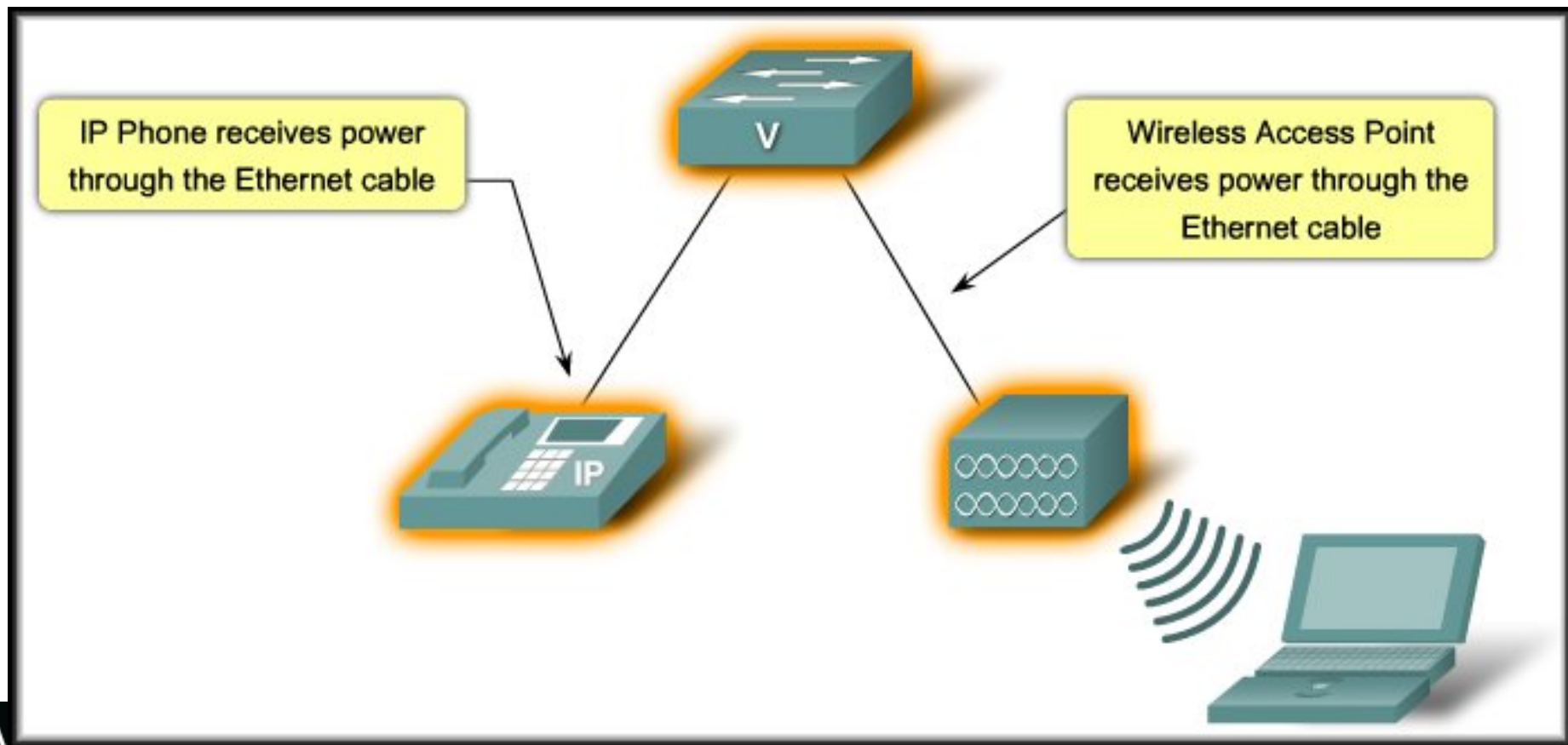




Switch Performance Characteristics

Power over Ethernet (PoE)

- Allows the switch to deliver power to a device over the existing Ethernet cabling.

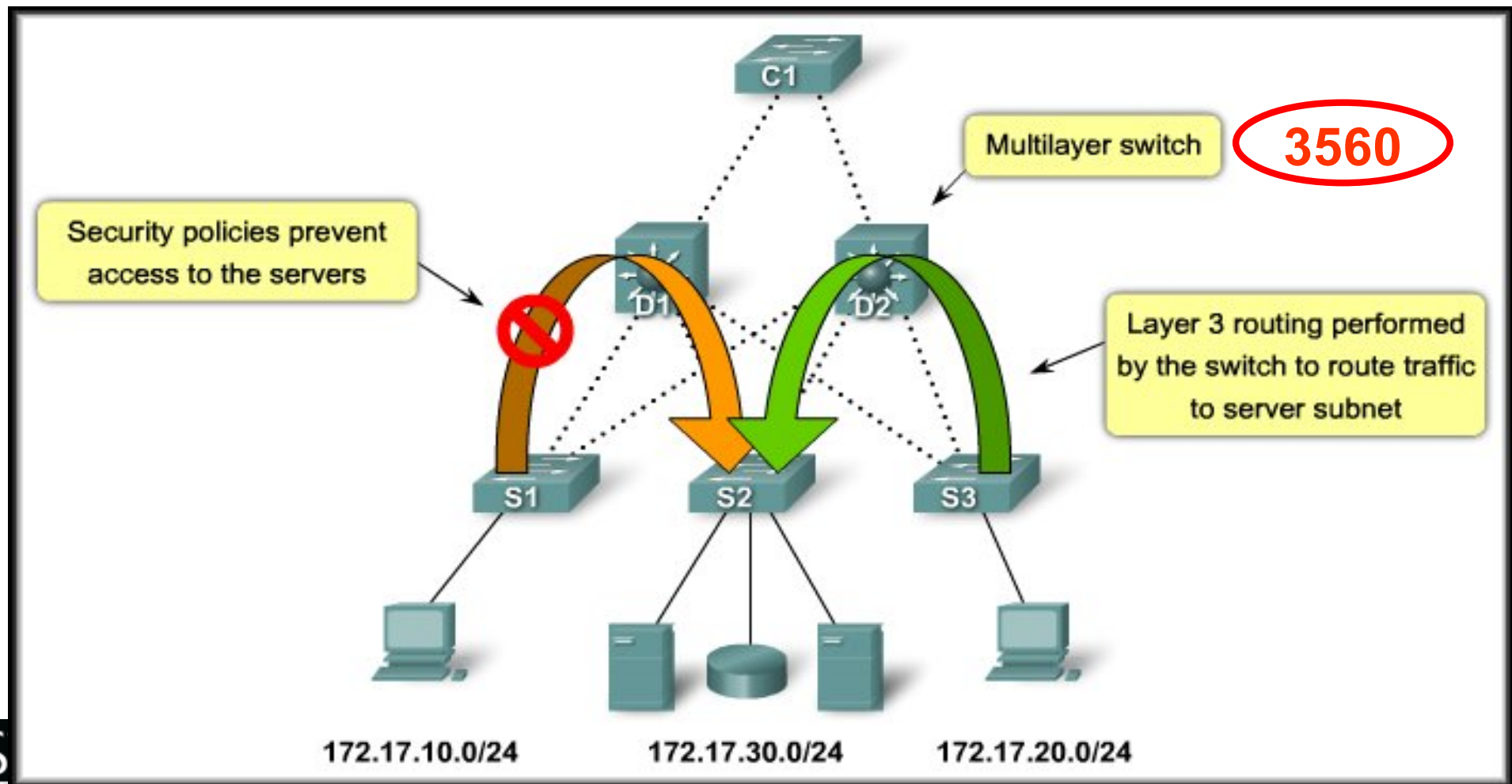




Switch Performance Characteristics

Layer 3 Functionality

Switches typically operate at Layer 2 of the OSI Model.

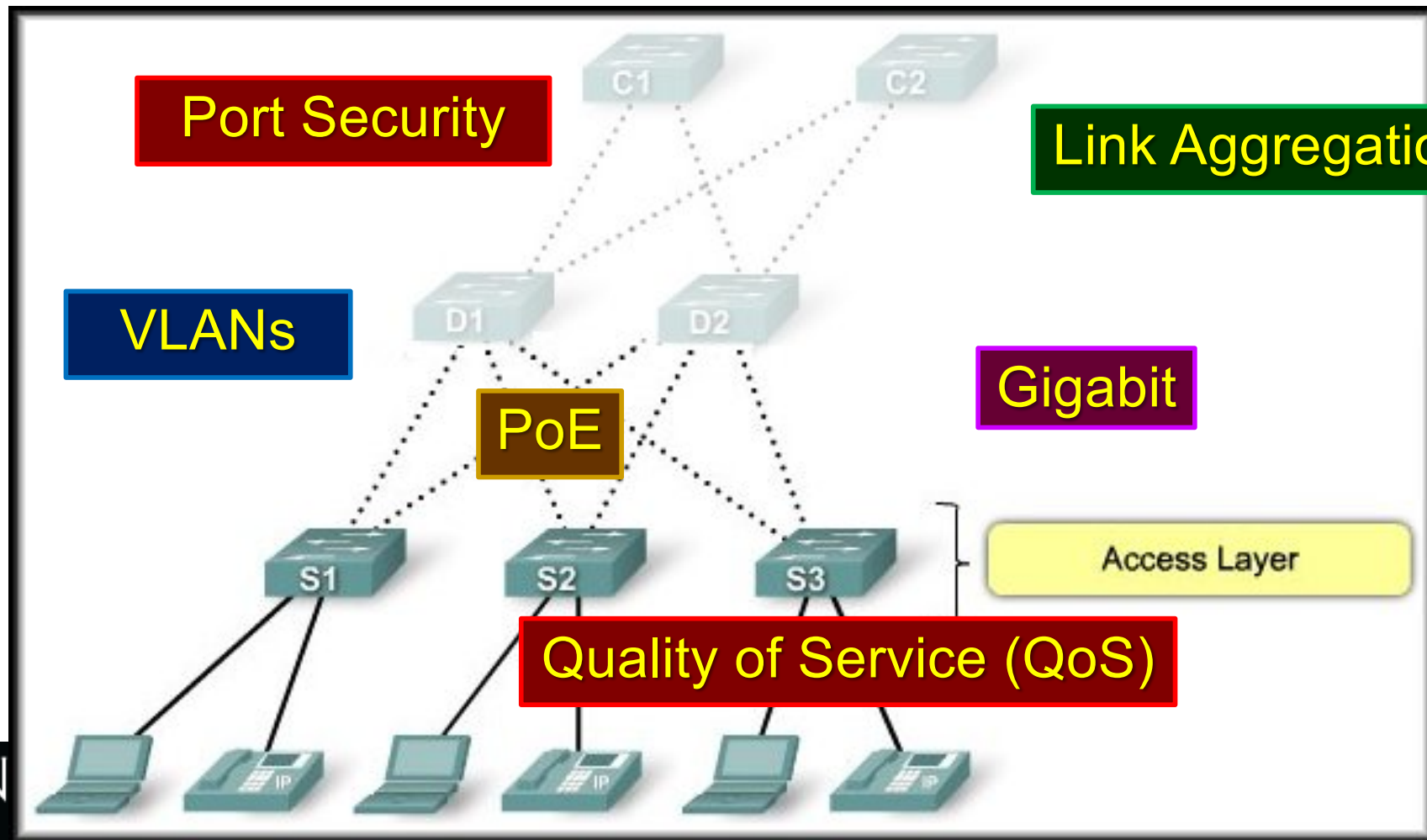




Switch Features

Hierarchical Network

Access Layer Switch Features:

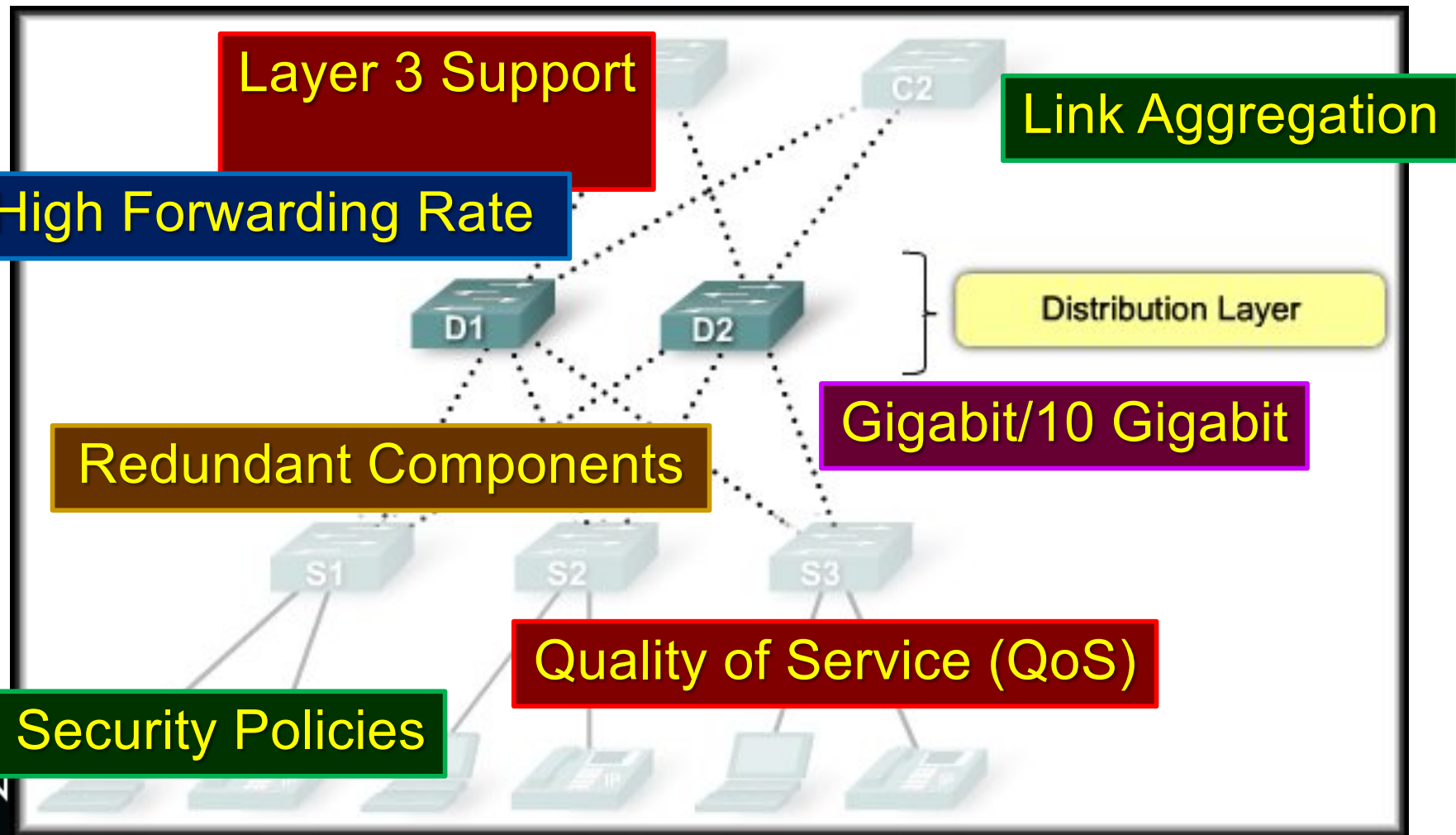




Switch Features

Hierarchical Network

- Distribution Layer Switch Features:





Switch Features

Hierarchical Network

Core Layer Switch Features:

Layer 3 Support

Very High Forwarding Rate

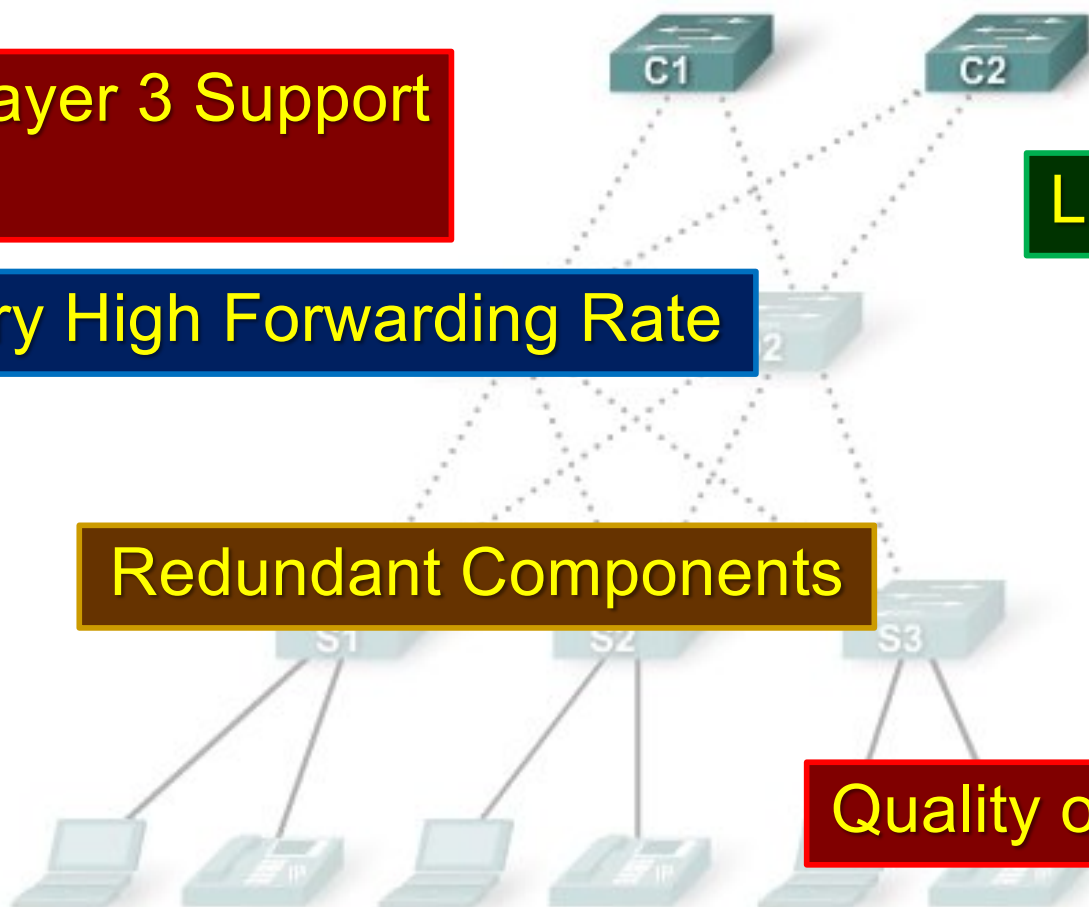
Redundant Components

Quality of Service (QoS)

Link Aggregation

10 Gigabit

Core Layer





LAN Design Summary

In this lecture, we covered:

- Hierarchical Network Model
- Converged Networks
- Network Design Considerations
- Switch Performance Characteristics
- Switch Features